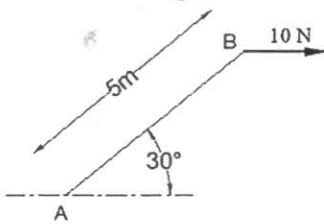
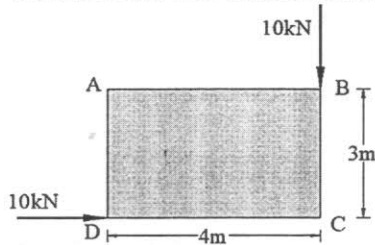


6. ΣF_x for the given force system is _____



- A) 0 N
B) 10 N
C) 19.66 N
D) 8.66 N
7. Determine the Moment at point C for the figure shown below.



- A) 10 kN-m
B) -30 kN-m
C) 0 kN-m
D) 8.66 kN-m
8. _____ is applied while trying to turn a key into a lock.
A) couple
B) non-coplanar force
C) momentum
D) coplanar force
9. A load which varies with the length of the beam is known as _____.
A) point load
B) uniformly distributed load
C) uniformly moving load
D) uniformly varying load
10. The equivalent point load for a UVL will be equal to _____.
A) area of the rectangle
B) area of the triangle
C) maximum magnitude of UVL
D) zero
11. The below figure depicts _____ beam.



- A) propped cantilever
B) continuous
C) fixed
D) cantilever
12. The below figure depicts _____ beam.
-
- A) fixed
B) continuous
C) propped cantilever
D) cantilever
13. The Friction experienced by a body when it is moving is called
A) Dynamic Friction
B) Sliding Friction
C) Static Friction
D) Rolling Friction
14. The magnitude of limiting friction bears a constant ratio to _____ is called coefficient of friction
A) Normal Reaction
B) Frictional Resistance
C) Wet Friction
D) Dry friction

Angle of friction is angle between -----

- A) The weight of the body and the friction here
 B) The incline and horizontal
 C) Normal reaction and the resultant
 D) The normal reaction and friction force
16. Frictional force is independent of
 A) Angle of repose
 B) Angle of friction
 C) Coefficient of friction
 D) Shape and size of surface of contact
17. Moment of inertia is a
 A) Second moment of an area
 B) Third moment of an area
 C) Fourth moment of an area
 D) Moment of an area
18. Moment of Inertia is the measure of
 A) resistance of the body against rotation
 B) resistance of the body against movement
 C) resistance of the body against compression
 D) resistance of the body against tension
19. The unit of Moment of Inertia in SI system is
 A) mm^3
 B) m^4
 C) mm^2
 D) mm
20. if the moment of inertia is more about a particular axis, then bending of the section would be -----about that axis
 A) zero
 B) easy
 C) no effect
 D) difficult

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain basic idealizations assumed while applying the principles of mechanics to solve the practical problems.	4	L*2	1	1,2
	b) Differentiate between coplanar concurrent and coplanar collinear force system.	4	L2	1	1,2
	c) Find the reactions at the contact points of two identical cylinders of weight 100N each placed in a trench as shown in Fig. Q 1 (c).				

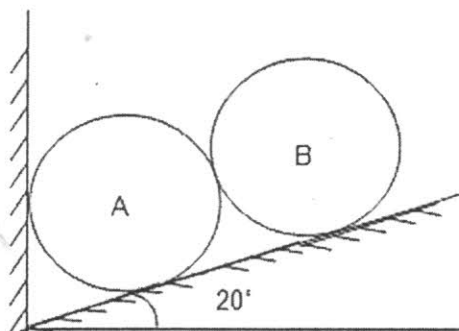


Fig. Q 1 (c)

		8	L3	2	1,2
2.	a) Define free body diagram with suitable example.	4	L2	2	1,2
	b) What is couple what are its characteristics.	4	L2	2	1,2

- c) Determine the magnitude and direction of the resultant force for the force system as shown in Fig. Q 2 (c).

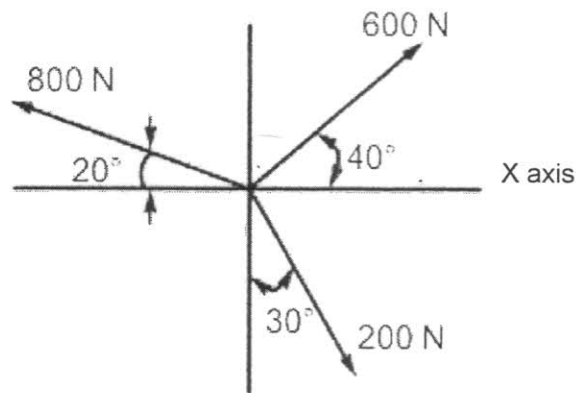


Fig. Q 2 (c)

8 L3 1 1,2

3. a) What is moment of a force what are its characteristics.
 b) Explain principle of transmissibility with neat sketch.
 c) Determine the resultant of the coplanar non concurrent force system with respect to point 'C' as shown in Fig. Q 3 (c).

4 L2 1 1,2

4 L2 1 1,2

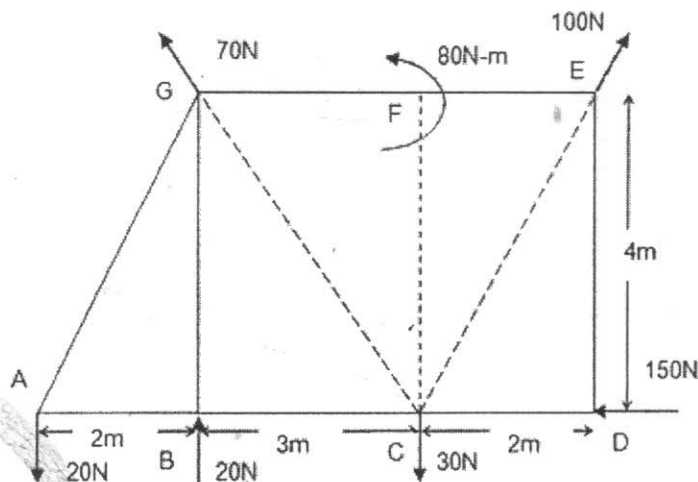


Fig. Q 3 (c)

8 L3 2 1,2

Unit – II

4. a) State Coulomb's laws of friction (any four).
 b) Define i) Angle of repose and ii) angle of friction
 c) Determine the support reactions for the overhanging beam shown in the Fig. Q 4 (c).

4 L2 4 1,2

4 L2 4 1,2

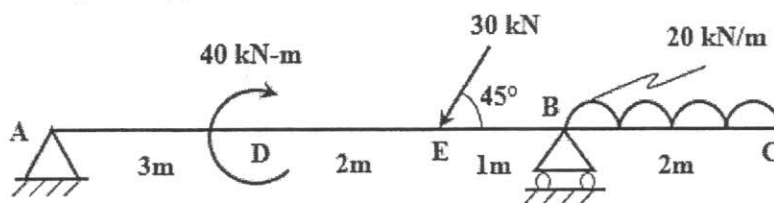


Fig. Q 4 (c)

8 L3 3 1,2

5. a) With examples explain the statically determinate and statically indeterminate beams.
 b) Explain cantilever and propped cantilever beam with neat sketch.
 c) A block of weight 20N is placed on an inclined plane as shown in the Fig. Q 5 (c). is subjected to a force P that is parallel to the plane. Taking the inclination of plane with respect to horizontal as 30° and the coefficient of friction as 0.24, determine the value of P for impending motion of the block towards ground.

4	L2	3	1,2
4	L2	3	1,2

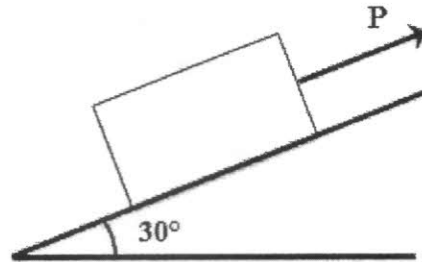


Fig. Q 5 (c)

8	L3	4	1,2
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6. a) Explain Uniformly distributed and Uniformly Varying load (UDL&UVL) with neat sketch.
 b) Explain any four types of supports with the neat sketches.
 c) A ladder 6m long having a weight of 150N is placed against a vertical wall so that it makes an angle of 60° with the ground. How far up the ladder can a 700N man climb before the ladder is on the verge of slipping? The coefficient of friction for all contact surface is 0.26.

4	L2	3	1,2
4	L2	3	1,2

8	L3	4	1,2
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Unit – III

7. a) State and prove the perpendicular axis theorem.
 b) Derive an expression for moment of inertia of a Triangular lamina about its base by using method of integration.
 c) Locate the centroid of the shaded area shown in the Fig. Q 7 (c) with respect to the given axis.

4	L1	5	1,2
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4	L2	5	1,2
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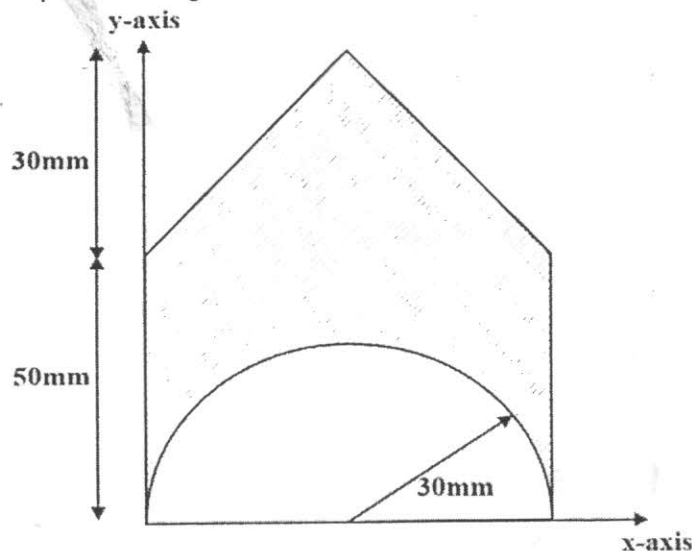


Fig. Q 7 (c)

8	L3	5	1,2
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8. a) Distinguish between centroid and center of gravity.
b) Derive an expression for centroid of a Quarter circle radius 'r' from the method of first principle.
c) Determine the moment of inertia for the section shown in the fig.Q.8 © about the axis passing through the centroid which is parallel to the topmost fiber of the section and determine the moment of inertia about the axis of symmetry.

4 L1 5 1,2

4 L2 5 1,2

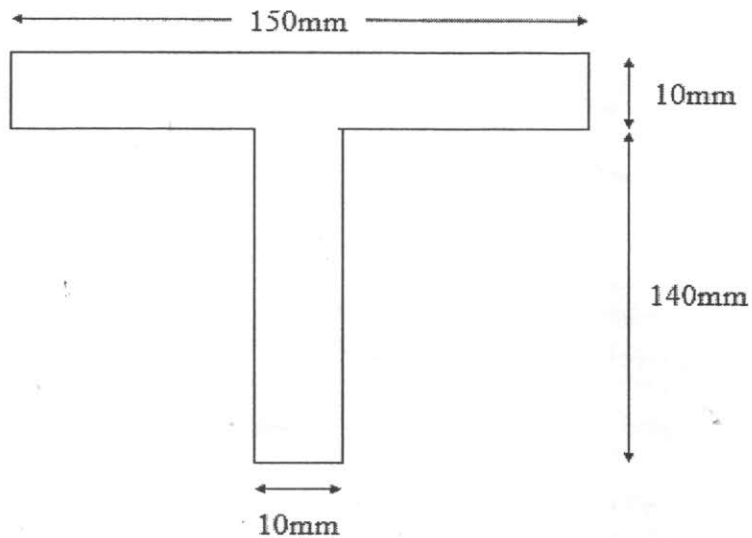


Fig. Q 8 (c)

8 L3 5 1,2